Fiscal Year:	FY 2012	**Task Last Updated:**	FY 02/19/2013
PI Name:	Bloomberg, Jacob J. Ph.D.		
Project Title:	Development of Countermeasures to Enhance Sensorimotor Adaptation		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Crew Health NNJ07ZSA002N
Start Date:	10/01/2008	**End Date:**	09/30/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Cohen, Helen (Baylor College of Medicine) Reschke, Millard (NASA Johnson Space Center) Mulavara, Ajitkumar (USRA) Buccello-Stout, Regina (Wyle Integrated Sciences and Engineering Group) Peters, Brian (Wyle Laboratories)		
Grant/Contract No.:	NCC 9-58-SA01601		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

During adaptation to novel gravitational environments, sensorimotor disturbances have the potential to disrupt the ability of astronauts to perform required mission tasks. The overall goal of this project is to gain the information necessary to develop a comprehensive sensorimotor adaptability (SA) training program to facilitate rapid adaptation to novel gravitational environments. It has been previously shown that subjects trained to adapt to varied sensorimotor challenges can adapt faster to new sensory environments that they have never experienced before. This is a process known as adaptive generalization and it allows you to enhance the ability to learn how to learn to adapt to novel environments. By applying these motor-learning concepts for training astronauts we can enhance their ability to rapidly adapt their behavioral responses following a gravitational transition. To minimize cost and demands on crew time we have integrated SA training with existing exercise activities, namely treadmill walking. The SA training program we are developing entails manipulating the sensory conditions of treadmill exercise to systematically challenge multiple sensorimotor systems while conducting nominal exercise activities. To provide SA training we have mounted a treadmill on a six degree-of-freedom motion base to produce variation in the support surface along with variation in visual input during walking using both a projected virtual scene that produces variation in visual flow or through goggles that distort visuomotor information.
Below is an overview of our research results to-date:

Study 1. Aim: Determine whether sensorimotor adaptability (SA) skills learned during treadmill SA training could transfer to a novel discordant sensory environment and to quantify the extent to which any training effects would be retained. This study showed that SA training enhances locomotor adaptability and increases multi-tasking capability during adaption to a novel discordant sensory environment. Results also demonstrated that this increased adaptability can be retained up to 6 months and perhaps longer. Thus SA training improves both locomotor function and multi-tasking capabilities which are essential components required to complete critical mission tasks.

Study 2. Aim: Determine whether performance in a novel discordant sensory environment can be predicted based on a subject's inherent visual dependency. This study showed that subjects with greater visual dependency have decreased locomotor stability and reduced ability to multi-task when negotiating novel sensory discordant conditions. These data indicate that visual dependency may be a marker for decreased ability to adapt to novel environments. Identifying preflight predictors of sensorimotor adaptability can be used to develop individualized training prescriptions that target the specific needs of each crewmember thus making the training process targeted and more efficient.

Study 3. Aim 1: Examine the influence of visual dependence on a subject's SA training performance. Aim 2: Determine whether visual dependence influences a person's ability to generalize newly acquired adaptive skills in unfamiliar discordant sensory conditions. This study demonstrated that highly visually dependent individuals receive the greatest benefit from SA training. Although both visually dependent and independent subjects perform equally well by the end of their training, the visually independent subjects outperformed their highly visually dependent counterparts on a locomotor test in a novel sensory discordant environment. This suggests that visually dependent subjects may receive the greatest benefit from SA training but they are less able to generalize their skills.

Study 4. Aim: Determine if SA training using treadmill walking transfers to other functional tasks including manual control. Subjects performed two operationally oriented manual control tests after locomotor SA training. Results indicate that training tasks need to contain at least some of the critical features of the criterion task to be effective. This result informs the design of future training programs ensuring that they contain sufficient locomotor and manual challenges to allow for a comprehensive training across performance modalities.

Study 5. Aim: The goal of this study was to quantify the adaptive locomotor effects, as well as the cognitive and metabolic costs, of exposure to a discordant sensory environment experienced during SA training. Greater metabolic cost incurred during balance instability means more physical work is required during adaptation to new environments possibly affecting crewmembers' ability to perform mission critical tasks during early surface operations on planetary expeditions. Results confirmed that walking in discordant conditions not only compromises locomotor stability and the ability to multi-task, but also increased metabolic cost. Importantly, like locomotor stability and multi-tasking ability, metabolic expenditure while walking in discordant sensory conditions improved during adaptation. This finding confirms that SA training can benefit multiple performance parameters central to the successful completion of critical mission tasks.

Study 6. Aim: The goal of this study was to determine whether performance improvements that result from SA training using visual and support surface discordance would improve performance during exposure to novel vestibular discordance produced by Galvanic Vestibular Stimulation (GVS). GVS has been previously shown to simulate the vestibular disturbances associated with space flight. Results show that SA training enhanced dual-tasking capability during initial exposure to GVS demonstrating that SA training improves performance during simulation of vestibular disturbances associated with exposure to space flight.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Sensorimotor adaptability training programs have Earthbound application in rehabilitation of patients with balance disorders, and for fall prevention training among seniors. The most important factors underlying morbidity in the older adult population are injurious falls and the restriction of activity due to falls. Approximately 25-35% of community-dwelling persons older than 65 years fall at least once a year, and approximately 40-50% of fallers experience two or more falls. The outcome is that 40% of all nursing home admissions are due to injuries due to falls. Age-related sensory changes and deterioration in ability to compensate in older adults can contribute to increase incidence of falling. We previously conducted a study to determine if balance training using variation in visual flow during treadmill exercise improves functional mobility in healthy older adults who were experiencing age-related postural instabilities (Buccello-Stout et al. 2008). The results showed that subjects who were exposed to varied visual flow during treadmill walking significantly improved their ability to negotiate an obstacle course after training compared to another elder group who only walked on a treadmill for the same amount of time. Importantly, the training benefit was retained when subjects were tested four weeks later. The training also allowed the elder subjects to adopt a head stabilization strategy optimized to maintain postural equilibrium control (Buccello-Stout et al., 2012). These studies confirm that adaptability training developed for use by astronauts can also be used to improve balance and gait performance in elder subjects and points to the general applicability of this type of training in different clinical populations.

Task Progress:	1) In FY12 we completed the data collection and analysis for fifth and sixth studies in this project. 2) During this review period four presentations related to this project were completed at four international scientific meetings. These include: Bloomberg JJ, Peters BT, Mulavara AP, Brady RA, Batson CD, Miller CA, Cohen HS. Countermeasures to enhance sensorimotor adaptability. Major Problems of Space Life Sciences Conference, Moscow, Russia, October 20-21, 2011. Bloomberg JJ, Peters BT, Mulvara AP, Brady RA, Batson CD, Miller CA, Ploutz-Snyder RJ, Guined JR, Buxton RE, Cohen HS. Locomotion adaptation improves balance control, multitasking ability and reduces the metabolic cost of postural instability. NASA Human Research Program Investigators' Workshop, February 14–16, 2012, Houston, TX. Mulavara AP, Wood SJ, Cohen HS, Bloomberg JJ. Locomotor dysfunction after long-duration space flight and development of countermeasures to facilitate faster recovery. 39th Scientific Committee on Space Research (COSPAR), Mysore, India, July 14-22, 2012. Bloomberg JJ, Mulavara AP, Peters BT, Wood SJ, Reschke MF. Sensorimotor countermeasures: Improving performance during gravitational transitions. 84rd Annual Scientific Meeting of the Aerospace Medical Association, Chicago Ill, May 12-16, 2013. 3) Seven manuscripts were published or are in press: Batson CD, Brady RA, Peters BT, Ploutz-Snyder RJ, Mulavara AP, Bloomberg JJ. Gait training improves performance in healthy adults exposed to novel sensory discordant conditions. Experimental Brain Research. 209(4): 515-24. 2011. Brady RA, Peters BT, Batson CD, Ploutz-Snyder RJ, Mulavara AP, Bloomberg JJ. Gait adaptability training is affected by visual dependency. Experimental Brain Research. 220: 1-9. 2012. Buccello-Stout RR, Cromwell RL, Bloomberg JJ, Whorton EB. Effects of sensorimotor adaptation training on head stability movement control in response to a lateral perturbation in older adults. The Journal of Aging and Physical Activity. In Press. Brady RA, Brian BT, Bloomberg JJ. Walking on an oscillating treadmill: Strategies of stride-time adaptation. Ecological Psychology. 24: 265-278. 2012. Mulavara AP, Ruttley T, Cohen HS, Peters BT, Miller C, Brady R, Merkle L, Bloomberg JJ. Vestibular-somatosensory convergence in head movement control during locomotion after long-duration space flight. Journal of Vestibular Research. 22(2):153-66. 2012. Cohen HS, Kimball KT, Wood SJ, Paloski WP, Bloomberg JJ, Mulavara AP. The role of posturography and locomotor testing in assessing performance in astronauts. Journal of Vestibular Research. 22(4): 191-6. 2012. Peters BT, Brady RA, Batson CD, Guined JR, Ploutz-Snyder RJ, Mulavara AP, Bloomberg JJ. Changes in locomotor stability, cognition, and metabolism during adaptation to walking in discordant sensory conditions. Aviation, Space, and Environmental Medicine, In Press.
Bibliography Type:	Description: (Last Updated: 11/14/2025)
Articles in Peer-reviewed Journals	Peters BT, Brady RA, Bloomberg JJ. "Walking on an oscillating treadmill: Strategies of stride-time adaptation." Ecological Psychology. 2012 Nov;24(4):265-78. http://dx.doi.org/10.1080/10407413.2012.702637 , Nov-2012
Articles in Peer-reviewed Journals	Brady RA, Peters BT, Batson CD, Ploutz-Snyder RJ, Mulavara AP, Bloomberg JJ. "Gait adaptability training is affected by visual dependency." Experimental Brain Research. 2012 Jul;220(1):1-9. PMID: 22585123 ; http://dx.doi.org/10.1007/s00221-012-3109-5 , Jul-2012
Articles in Peer-reviewed Journals	Cohen HS, Kimball KT, Mulavara AP, Bloomberg JJ, Paloski WH. "Posturography and locomotor tests of dynamic balance after long-duration spaceflight." Journal of Vestibular Research. 2012;22(4):191-6. http://dx.doi.org/10.3233/VES-2012-0456 ; PubMed PMID: 23142833 , Jan-2012
Articles in Peer-reviewed Journals	Mulavara AP, Ruttley T, Cohen HS, Peters BT, Miller C, Brady R, Merkle L, Bloomberg JJ. "Vestibular-somatosensory convergence in head movement control during locomotion after long-duration space flight." Journal of Vestibular Research. 2012 Jan;22(2):153-66. PMID: 23000615 ; http://dx.doi.org/10.3233/VES-2011-0435 , Jan-2012
Articles in Peer-reviewed Journals	Peters BT, Brady RA, Batson CD, Guined JR, Ploutz-Snyder RJ, Mulavara AP, Bloomberg JJ. "Adaptation in locomotor stability, cognition, and metabolic cost." Aviation, Space, and Environmental Medicine. 2013 Jun;84(6):567-72. PubMed PMID: 23745284 (originally reported as 2013. In Press. Feb-2013) , Jun-2013
Articles in Peer-reviewed Journals	Buccello-Stout RR, Cromwell RL, Bloomberg JJ, Whorton EB. "Sensorimotor adaptation training's effect on head stabilization in response to a lateral perturbation in older adults." Journal of Aging and Physical Activity. 2013 Jul;21(3):272-89. Epub 2012 Sep 18. PubMed PMID: 23006335 (originally reported as 2012 Sep 18. [Epub ahead of print]) , Jul-2013
Articles in Peer-reviewed Journals	Macaulay TR, Peters BT, Wood SJ, Clément GR, Oddsson L, Bloomberg JJ. "Developing proprioceptive countermeasures to mitigate postural and locomotor control deficits after long-duration spaceflight." Front Syst Neurosci. 2021 Apr 27;15:658985. Review. https://doi.org/10.3389/fnsys.2021.658985 ; PMID: 33986648; PMCID: PMC8111171 , Apr-2021
Awards	Mulavara A. "NASA Tech Brief Award, October 2012." Oct-2012
Awards	Bloomberg J. "NASA Performance Award, July 2012." Jul-2012
Awards	Bloomberg J. "NASA Tech Brief Award, October 2012." Oct-2012